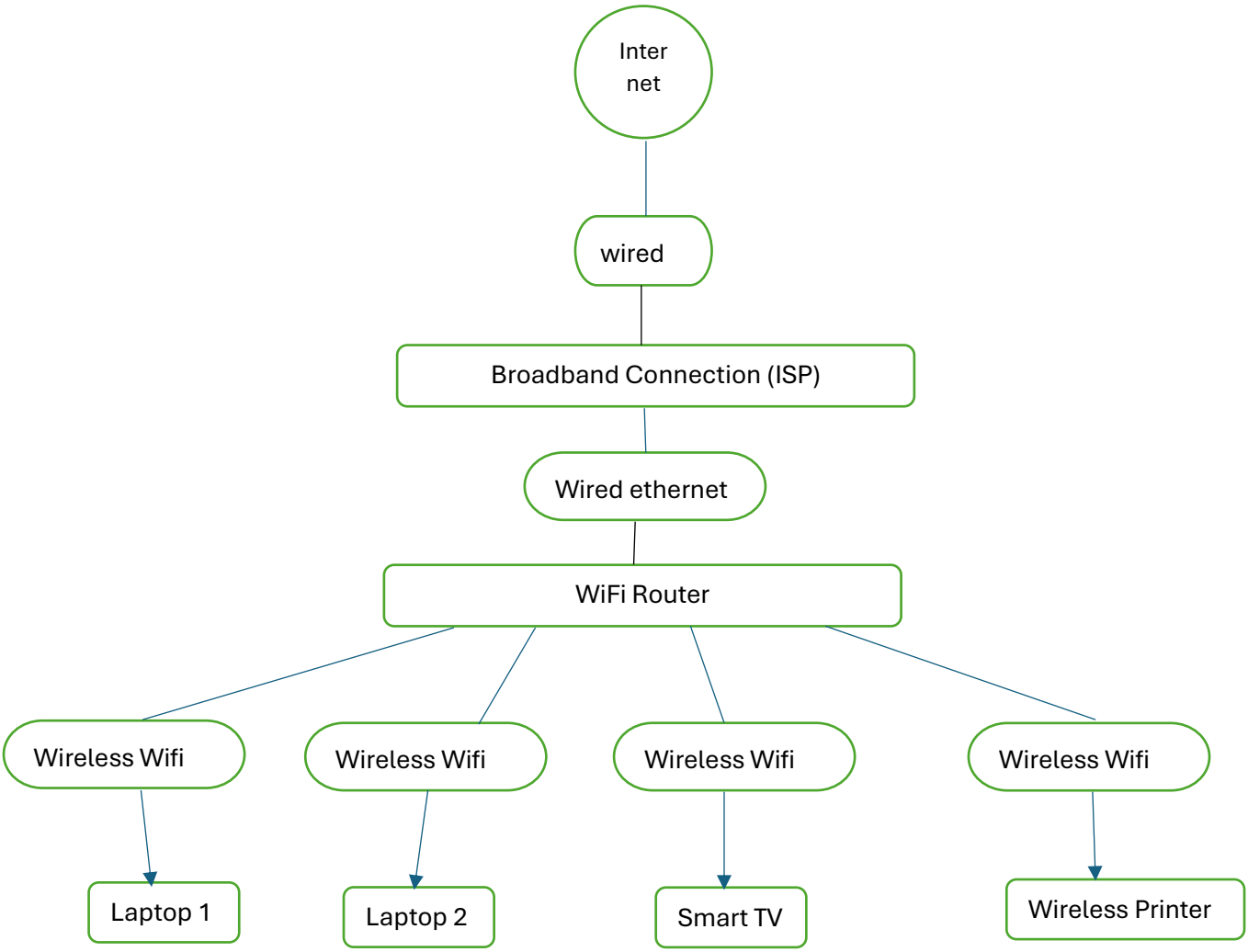
1. Draw a network diagram for a small office/home office (SOHO) setup that includes: one broadband internet connection, one Wi-Fi router, two laptops, one smart TV, and one printer. Clearly label each device and connection type (wired or wireless).



2. List all the hardware components (with brand/model suggestions) you would need to physically build the SOHO network diagrammed above, including cables and any switches or extenders if needed.

Hint: Check current listings on Flipkart or Amazon for typical device models used in Indian homes/offices.

SOHO Network: Complete Hardware Requirements

For this setup, you need one broadband connection, one Wi-Fi router, two laptops, one Smart TV and one wireless printer.

Below is a practical shopping list using brands and models commonly used in Indian homes and small offices.

No.	Hardware component	Quantity	Suggested brand/model	Purpose
1	Broadband internet connection	1	JioFiber, Airtel Xstream Fiber, or BSNL FTTH	Provides internet access.

No.	Hardware component	Quantity	Suggested brand/model	Purpose
2	Wi-Fi router	1	TP-Link Archer C6 AC1200	Connects devices to the network through wired and wireless connections.
3	Laptop	2	HP, Dell, or Lenovo	Used for office work, browsing, and other tasks.
4	Smart TV	1	Xiaomi Smart TV or Samsung Smart TV	Streams online videos and other internet content.
5	Wireless printer	1	Brother DCP-T426W	Prints documents over Wi-Fi.
6	Cat6 Ethernet cable	1–2	Zebronics or D-Link Cat6	Connects the broadband equipment to the router and, if desired, the printer or TV to the router.
7	Router power adapter	1	Supplied with the router	Provides power to the router.
8	Broadband modem/ONT	1, if required	Supplied by the internet service provider	Converts the incoming broadband connection into an Ethernet connection for the router.
9	Power extension board	1, if required	Anchor or GM	Provides additional power sockets for networking equipment.

3. Write step-by-step instructions to configure the Wi-Fi router for your SOHO network, including setting up a unique SSID, enabling WPA2 security, and changing the default admin password.

Step 1: Connect the Router

1. Connect the broadband modem/ONT to the router's **WAN/Internet port** using an Ethernet cable.
2. Connect the router's power adapter and switch it on.
3. Wait for the router's indicator lights to stabilize.
4. Connect your laptop to the router using Wi-Fi or an Ethernet cable.

Step 2: Open the Router Administration Page

1. Open a web browser on your laptop.
2. Enter the router's management address in the address bar. Common addresses include:
    - 192.168.0.1
    - 192.168.1.1
    - tplinkwifi.net
3. Press **Enter**.

4. Log in using the router's existing administrator credentials, or follow the initial setup instructions if the router is new.

Step 3: Change the Default Administrator Password

1. Open **Advanced** or **System Tools** in the router settings.
2. Find **Administration**, **Password**, or **Change Password**.
3. Enter the current administrator password.
4. Create a strong new password.
5. Confirm the new password.
6. Click **Save** or **Apply**.

Step 4: Set a Unique Wi-Fi Network Name (SSID)

1. Open **Wireless Settings** or **Wireless → Wireless Settings**.
2. Find the **Network Name (SSID)** field.
3. Enter a unique network name.

Example:

HomeOffice_Network

4. If your router supports both 2.4 GHz and 5 GHz bands, configure the SSID for each band. You can use the same name for both or separate names if you prefer.
5. Click **Save** or **Apply**.

Step 5: Enable WPA2 Wi-Fi Security

1. Open **Wireless Security** or the Wi-Fi security settings.
2. Select the appropriate Wi-Fi band.
3. Find the **Security** or **Authentication Type** option.
4. Select **WPA2-PSK (AES)** or **WPA2-Personal (AES)**.
5. Enter a strong Wi-Fi password in the **Wireless Password** or **Pre-Shared Key** field.
6. Click **Save** or **Apply**.
7. Repeat these steps for the other Wi-Fi band if necessary.

Step 6: Reconnect Your Devices

After saving the settings, your Wi-Fi connection may disconnect.

1. Open the Wi-Fi settings on Laptop 1.
2. Select HomeOffice_Network.
3. Enter your new Wi-Fi password.
4. Repeat the process on Laptop 2.
5. Connect the smart TV to the same Wi-Fi network.
6. Connect the printer to the same Wi-Fi network using its wireless setup menu or the manufacturer's setup application.

Ensure all devices use the correct SSID and password.

Step 7: Test the Network

1. Connect both laptops to the Wi-Fi network.
2. Open a web browser on each laptop.
3. Visit a website to verify internet connectivity.
4. Check that the smart TV can access an online streaming service.
5. Print a test page from one of the laptops.
6. Confirm that you can still access the router administration page using your new administrator password.

4. Configure IP addresses for all devices in your SOHO network: assign static IPs to the printer and smart TV, and use DHCP for the laptops. Show a table listing each device, its MAC address (make one up), and the assigned IP address.

To configure IP addresses for your SOHO network, you will need to allocate static IP addresses outside your dynamic DHCP pool to prevent address conflicts. Below is a structured layout for your devices using a standard 192.168.1.X private network subnet.

SOHO Network Device IP Assignment

Device	MAC Address	IP Address	Assignment Type
Main Router (Gateway)	00:11:22:33:44:55	192.168.1.1	**Static (Default)**
Office Printer	00:11:22:AA:BB:CC	192.168.1.10	**Static**
Smart TV	00:11:22:DD:EE:FF	192.168.1.20	**Static**
Laptop 1	00:11:22:66:77:88	192.168.1.100	**DHCP (Dynamic)**
Laptop 2	00:11:22:99:00:AA	192.168.1.101	**DHCP (Dynamic)**

Configuration Steps

1. **Configure the DHCP Pool:** Log into your router admin panel and set the DHCP range to start at 192.168.1.100 and end at 192.168.1.254. This reserves IP addresses 192.168.1.2 through 192.168.1.99 for static assignments.
2. **Assign Printer & Smart TV IPs:** You can set these static IPs by either navigating to the **IP Reservation / DHCP Reservation** tab in the router settings and mapping their MAC addresses to the chosen IPs, or by manually typing the IP address, subnet mask (255.255.255.0), and gateway (192.168.1.1) directly into the network settings menu of each device.
3. **Configure Laptops:** Ensure both laptops have their IPv4 network settings set to **Obtain an IP address automatically (DHCP)** so they dynamically pull an available address from the pool when they connect.

5. Explain how you would troubleshoot if one of the laptops cannot connect to the internet in your SOHO network. List at least 3 possible causes and what steps you would take to diagnose and fix each issue.

1. IP Address Conflict or DHCP Lease Failure

- **Cause:** The laptop failed to pull a valid IP address from the router, or another device on the network is accidentally using the exact same IP address.
- **Diagnosis:** Open the command prompt (Windows: cmd, Mac/Linux: Terminal) and type ipconfig (Windows) or ifconfig (Mac/Linux). Check if the IPv4 address starts with 169.254.x.x (an APIPA address, meaning it failed to reach the DHCP server) or if a windows popup warns about an IP conflict.
- **Fix: Force the laptop to request a fresh IP address from the router.**
    - **Windows:** Type ipconfig /release and then ipconfig /renew.
    - **Mac/Linux:** Toggle the Wi-Fi card off and back on, or renew the DHCP lease in System Settings.

2. Misconfigured DNS Settings

- **Cause:** The laptop is connected to the router and has an IP address, but it cannot translate web domain names (like google.com) into IP addresses.
- **Diagnosis**: Ping a live external IP address directly by running ping 8.8.8.8. If the ping succeeds, but searching for "google.com" fails in your browser, your DNS configuration is broken.
- **Fix:** Manually assign reliable public DNS servers. Go to the laptop's network adapter properties and change the DNS settings from automatic to manual. Input 8.8.8.8 (Google DNS) as the Primary DNS and 1.1.1.1 (Cloudflare DNS) as the Secondary DNS.

3. Outdated or Corrupted Network Adapter Driver

- **Cause:** The physical Wi-Fi or Ethernet network interface card (NIC) software on the laptop has crashed, corrupted, or become incompatible.
- **Diagnosis:** Check the device manager settings. If the network adapter shows a yellow warning triangle, or if the Wi-Fi option completely disappears from your operating system menu, the driver is malfunctioning.
- **Fix: Perform a network reset or reinstall the driver.**
    - **Quick Fix:** Right-click the network adapter in Device Manager, select Disable, wait 10 seconds, and select Enable.
    - **Full Fix:** If that fails, select Uninstall Device, restart the laptop, and let the operating system automatically reinstall the default driver on boot.